Lesson 7: Capacitors

ChatGPT edition — isolated 3 V test

Charge, voltage, time, and stored energy

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

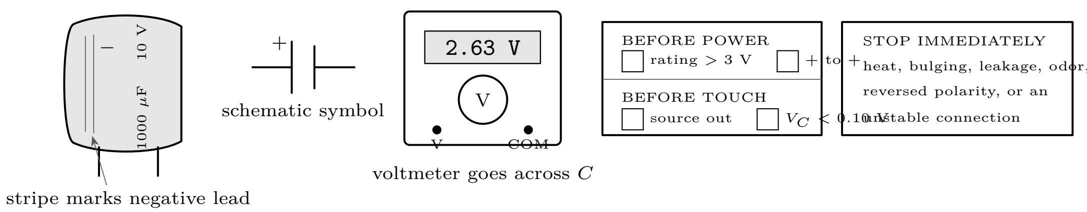
- Switched holder with 2 AA cells
- 1000 μF electrolytic capacitor, rated at least 10 V
- 10 kΩ and 22 kΩ resistors
- Digital multimeter and insulated leads
- Stopwatch, graph paper, removable labels

Predict first

Will capacitor voltage reach the measured source voltage at once? Sketch the shape you expect for charge and discharge; mark which part should change fastest.

Read the apparatus before connecting it

POLARITY, RATING, AND METER PLACEMENT

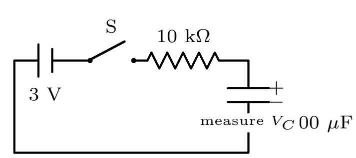


ADULT INSPECTION 1 — NO BATTERY INSTALLED

Read the capacitor body, identify its negative stripe and longer positive lead, and label both circuit nodes. Set the meter to DC volts with its black lead in COM. The adult checks the 10 kΩ resistor and every connection. **Never short a capacitor with metal.** Discharge only through a resistor.

Stage 1 — establish endpoints and polarity

1. With the capacitor absent, measure the two-cell source: $V_S =$ ______ V. Remove the battery holder.
2. Build the pictured circuit with the switch open. Red probe goes to the capacitor + node; black probe goes to its − node.
3. Ask before closing: *At $t = 0$, should V_C be nearer 0 V or V_S? What should the sign on the meter be?*
4. Adult installs the holder. Close S for 60 s. Record the final value $V_C =$ ______ V. It should be positive and approach V_S.
5. Open S and remove the holder. Keep the resistor in the path until the meter reads below 0.10 V. Confirm: ☐ discharged.



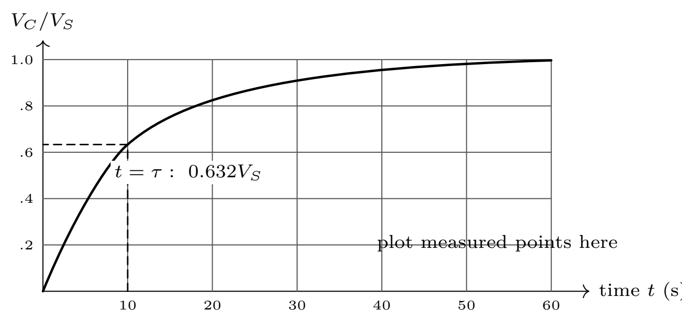
Stage 2 — measure a complete charge curve

ADULT INSPECTION 2

Capacitor reads below 0.10 V; polarity is unchanged; switch is open; probes are secure; source is still isolated 3 V. One person calls time while another reads.

1. Copy measured V_S into the table. At “start,” close S and keep it closed. Record V_C at every listed time; do not reposition clips live.
2. Between readings ask: *Are equal time intervals producing equal voltage increases? When is the slope steepest?*
3. At 50 s open S and remove the battery holder. Do not reverse the capacitor.

t (s)	0	2	5	10	20	30	40	50
V_C (V)	____	____	____	____	____	____	____	____
V_C/V_S	____	____	____	____	____	____	____	____



Worked time constant — calculate before judging the data

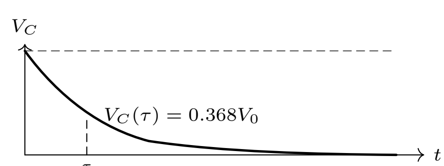
$$\tau = RC = (10{,}000\ \Omega)(0.001\ \mathrm{F}) = 10\ \mathrm{s}. \qquad V_C(\tau) = V_S(1 - e^{-1}) \approx 0.632V_S.$$

If $V_S = 3.00$ V, the 10 s target is 1.90 V. At $5\tau = 50$ s, $V_C \approx 0.993V_S$, close to the endpoint but never an instantaneous jump. Measured 10 s value: ______ V. Difference from target: ______ V.

Stage 3 — discharge through the same resistor

1. Begin near V_S, then remove the battery. Connect the 10 kΩ resistor across the capacitor; start time when the discharge path closes.
2. Record without touching conductors. Ask: *Does polarity reverse, or does the same positive voltage shrink toward zero?*

t (s)	0	5	10	20	30	40	50
V_C (V)	____	____	____	____	____	____	____



Stage 4 — change one thing

Replace only the 10 kΩ resistor with 22 kΩ. Keep the same capacitor, source, polarity, timing method, and voltage checkpoints.

Test	R	calculated τ	V_S	$V_C(\tau)$	time to $0.63V_S$	time to $0.90V_S$	result
A	10 kΩ	10 s					
B	22 kΩ	______					

Predict first

If resistance is multiplied by 2.2, what should happen to the time required to reach the same fraction of V_S? Prediction: ____

Diagnose before repeating

Observation	Check with power removed	Likely correction
Meter shows a minus sign	Probe colors and capacitor stripe	Put red on the labeled + node; do not reverse the capacitor to hide a probe error.
Voltage stays near 0 V	Switch continuity, lead seating, resistor value	Restore the intended series path; never bypass the resistor.
Voltage begins above 0.10 V	Residual charge	Battery out; discharge through 10 kΩ; measure again before touch.
Rise is much too fast or slow	Resistor bands and capacitor marking	Use measured R, stated C, and recompute RC.
Readings jump when untouched	Loose clips or changing contact	Open switch, remove battery, secure leads, then restart the whole timed run.
Heat, odor, leakage, or bulging	None—stop	Adult isolates the holder and retires the capacitor. Do not reuse it.

Explain from the evidence

Capacitance is $C = Q/V$. During charge, the initially large difference $V_S - V_C$ drives current through R; as V_C rises, that difference and the current shrink. During discharge, V_C itself drives current through the resistor, so the voltage falls exponentially without reversing polarity:

$$V_{\text{charge}} = V_S(1 - e^{-t/RC}), \qquad V_{\text{discharge}} = V_0 e^{-t/RC}.$$

Ideal stored energy is $E = \frac{1}{2}CV^2$. For 1000 μF at 3.00 V, $E = 0.0045$ J. The small value does not cancel the safe habit: treat every capacitor as charged until it is resistor-discharged and measured.

Final proof — prediction, pattern, control, conclusion

1. **Endpoint proof:** $V_C(0) \approx$ ______ V and $V_C(5\tau) \approx$ ______ V, not an instant jump.
2. **Shape proof:** equal time steps produced ☐ equal voltage changes ☐ progressively smaller changes.
3. **Time-constant proof:** at $t = RC$, charge was ______V_S and discharge was ______V_0.
4. **Controlled-change proof:** $22/10 = 2.2$; the measured time ratio was ______. This ☐ supports ☐ does not support $\tau \propto R$.
5. **Polarity proof:** during discharge the sign ☐ stayed positive ☐ reversed.

Conclusion: ____

FINAL ZERO-ENERGY CHECK

Switch open; battery holder removed; resistor remains across the capacitor; meter reads $V_C =$ ______ V, below 0.10 V. Adult initials: ______. Only then may the learner change or store components.